



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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ASSIGNMENT-One Page Writeup

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Guided by: Dr K Amandeep Singh

Course code/name: MLA0407-Deep Learning

Deep Learning-Based Recommendation System for Personalized Content

My primary responsibilities for this project focused on the foundational engineering, data pipeline development, and core deep learning architecture design.

- **Data Pipeline & Preprocessing:** I built a secure, reproducible ingestion workflow via the Kaggle API in Google Colab, preprocessing the pediatric X-rays by standardizing them to 224x224 pixels, normalizing pixel values to [0, 1], and applying real-time data augmentation (horizontal flips, zooming, shearing) to prevent overfitting.

CNN Architecture & Training Strategy: I designed a custom CNN with three convolutional blocks, ReLU activations, max-pooling, a 50% dropout layer, and a sigmoid output. Using the Adam optimizer, Binary Cross-Entropy loss, and an Early Stopping callback, the model successfully exceeded the 85% accuracy benchmark while controlling overfitting.

(Faculty signature)

T Bhupal

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